CHIP TO CROP 2026 · PROBLEM STATEMENT 04

AI That Investigates What Leaves Reveal

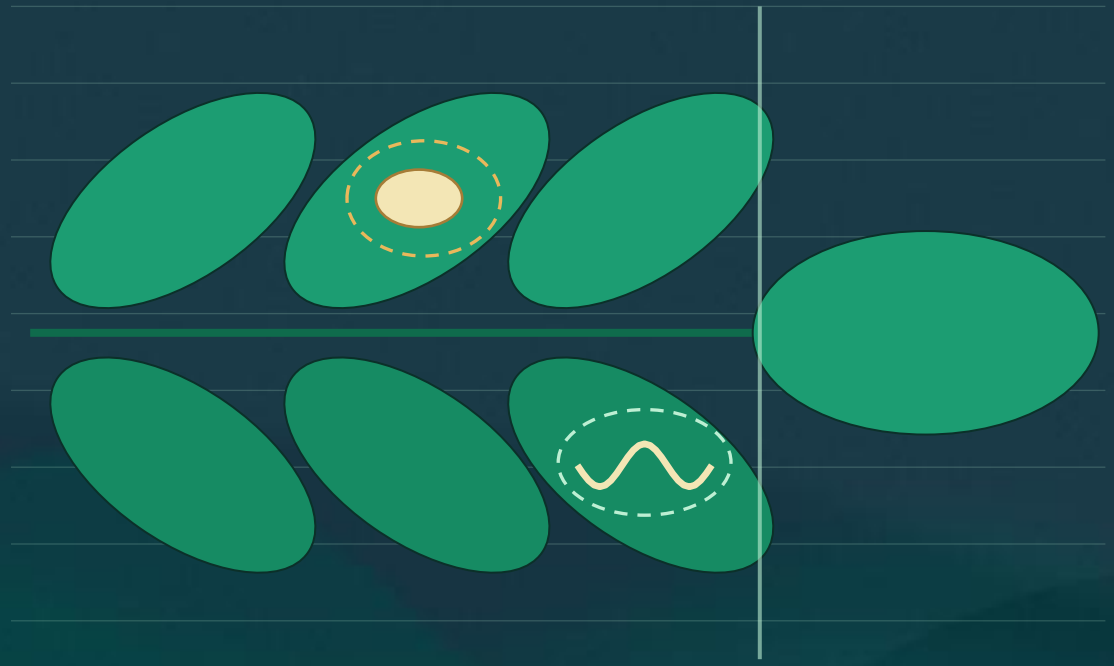
PhytoTrace — Explainable AI for Mixed Leaf-Miner Infestation Detection, Damage Assessment and Progression Monitoring in Tomato Crops

Trace the Damage. Reveal the Threat. Protect the Crop.

Team Name: Binary Bengaluru

Team Leader: Muskan Singh

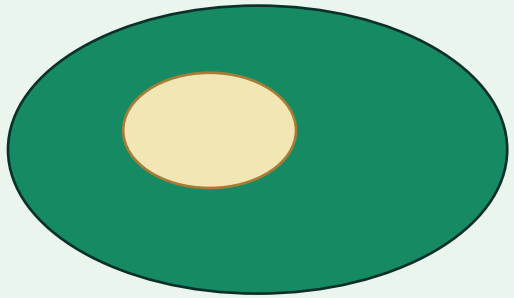
Akash · Jeevith · Hitesh · Mahaida



Illustrative concept visual — vector schematic, not a photograph

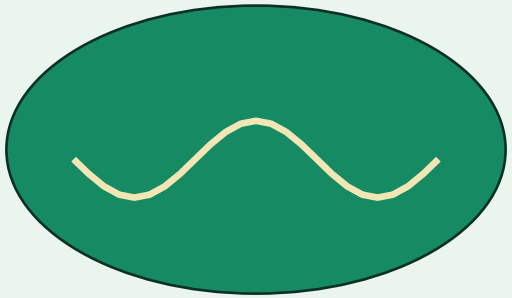
Two Pests. Similar Symptoms. One Critical Diagnostic Challenge.

Tuta-type mine



Irregular, blotch-like galleries (schematic)

Liriomyza-type mine



Narrow, winding serpentine galleries (schematic)

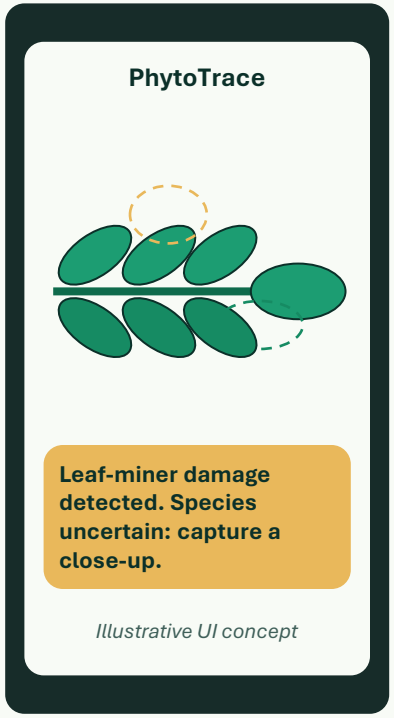
- *Tuta absoluta* and *Liriomyza* spp. are important leaf-mining pests of tomato
- Their visible damage can overlap
- Mixed infestation adds further diagnostic complexity
- One image-level label may be insufficient
- Farmers need understandable, traceable evidence

Schematic illustrations, not photographs. Real mines can look alike or overlap.

Overlapping symptoms → **Single-label prediction** → **Hidden mixed infestation** → **Unclear evidence trail** → **Harder scouting decisions**

From a Leaf Photograph to Actionable Evidence

1 Capture → 2 Quality check → 3 Detect mining damage → 4 Localize regions → 5 Analyze pest evidence → 6 Flag uncertainty or mixed → 7 Save and report



Quality feedback

Blur, glare and distance flagged before analysis

Highlighted regions

Suspected mines outlined with evidence labels

Plain-language message

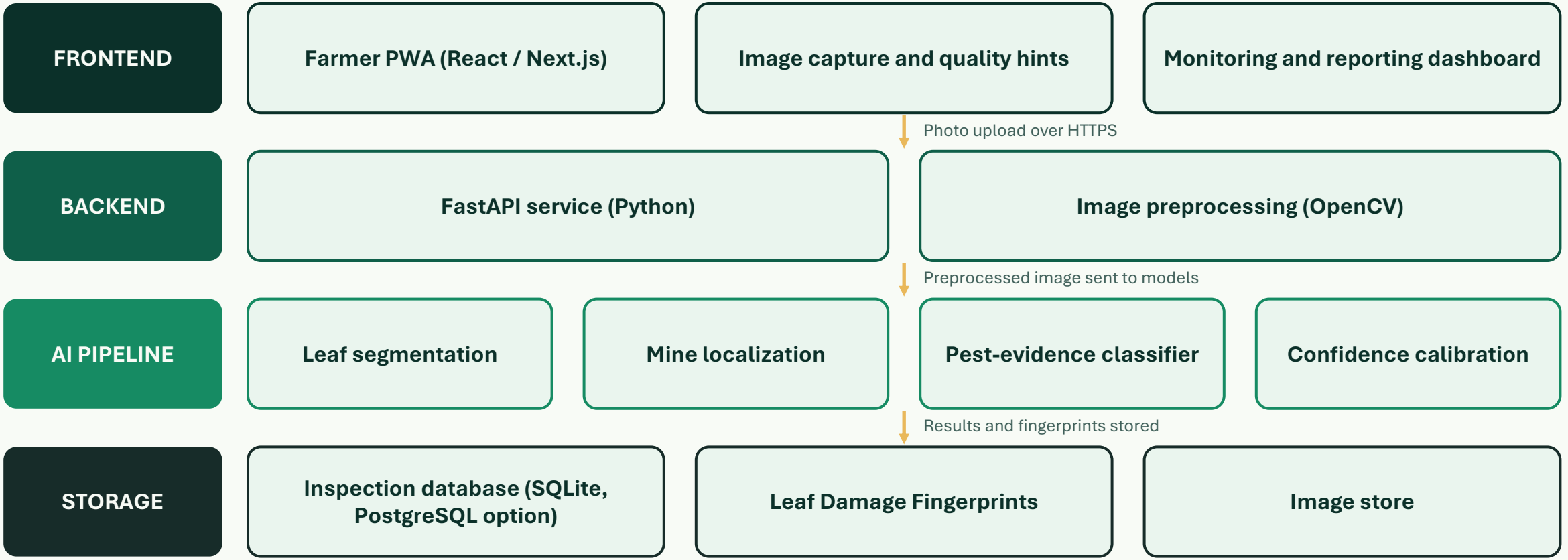
States what is known, what is uncertain, and the next step

Saved inspection

Damage Fingerprint and report linked to plant and field

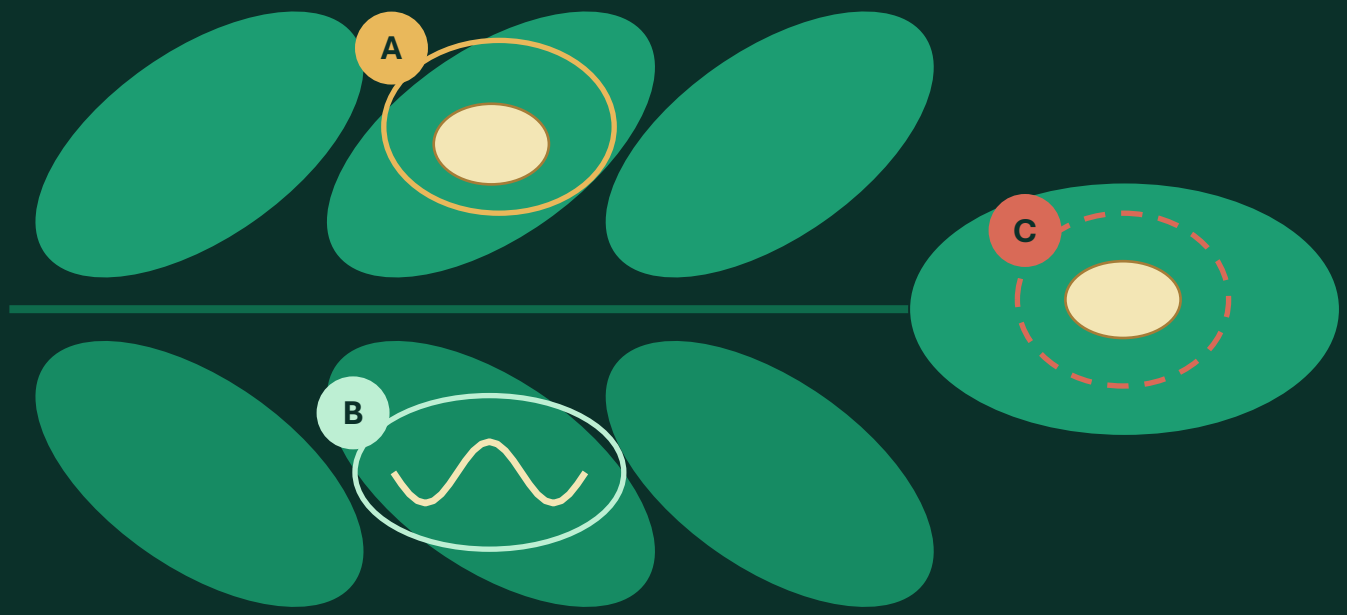
Proposed output — illustrative UI concept, not a screenshot of a trained system

A Smartphone-First AI Architecture



Future pathway (not yet implemented): export validated models to ONNX Runtime for on-device or offline inference; an Android/iOS wrapper is optional, not required.

The Mixed-Infestation Evidence Map



Illustrative UI concept — not the output of a trained model

A Tuta-consistent evidence

Broad, blotch-like mine morphology

B Liriomyza-consistent evidence

Narrow, winding serpentine path

C Uncertain evidence

Ambiguous cues: request close-up or expert review

Possible mixed infestation flagged: evidence, not a confirmed diagnosis

Pathway: close-up capture → expert verification → confirmed diagnosis

Overlays show where evidence sits, not why the model decided.

Beyond Classification: Six Layers of Intelligence

1 Regional evidence

Multi-label evidence for each damage region

2 Explainable localization

Overlays and region boundaries

3 Uncertainty-aware

Calibrated confidence and abstention

4 Damage Fingerprint

Structured record for every inspection

5 Longitudinal monitoring

Repeat inspections; visible-damage trend

6 Farm-level intelligence

Plant grid; unsampled plants are unknown

Capability	Basic classifier	PhytoTrace
Output unit	One label per image	Evidence per region
Explanation	Often none	Localization overlay
Uncertainty	Forced class	Can abstain or flag
Record	Label only	Damage Fingerprint
Over time	Single image	Repeat inspections
Scope	One leaf	Plant and field grid

Differentiation is the integrated use of these layers. Other tools may offer some of them individually.

From Research Prototype to Field-Ready Platform

1

Phase 1

Dataset research, licensing and annotation

2

Phase 2

Baseline detection and segmentation

3

Phase 3

Multi-label evidence and uncertainty handling

4

Phase 4

PWA and backend development

5

Phase 5

Inspection history and field analytics

6

Phase 6

Offline optimization and multilingual support

7

Phase 7

Expert validation and pilot testing

Most important technical dependency

Expert-confirmed Liriomyza and genuine mixed-infestation examples

Current status: Round 1 proposal (concept, architecture and validation plan). No trained model or field results are claimed.

Scientific Credibility Is Our Competitive Advantage

Metric	Question it answers	Protocol	Status
Precision, recall, F1	Are pest-related predictions correct and complete?	Per class, on held-out expert-labelled images	**Planned**
Mixed-infestation recall	Do we catch genuine mixed cases?	Expert-confirmed mixed samples	**Planned**
Segmentation IoU / Dice	Are mines and leaf area outlined accurately?	Pixel masks vs expert annotation	**Planned**
False positives	Are healthy or unrelated-damage leaves flagged?	Healthy and other-damage negatives	**Planned**
Confidence calibration	Does confidence match correctness?	Reliability diagram, ECE	**Planned**
Unknown-case rejection	Does it abstain on out-of-scope input?	Held-out unseen conditions	**Planned**
Inference latency	Is it fast enough in the field?	Capture-to-result time on target phones	**Planned**
Usability and field testing	Can farmers and scouts use it?	Scouting trials with users and experts	**Planned**

Splits separated by plant, farm or acquisition session to prevent leakage. No results are reported in this proposal.

Designed for Farmers. Built for Agricultural Intelligence.

Smartphone access

Runs in a browser on phones, tablets and laptops

Repeatable scouting

Same plant, same protocol, over time

Evidence for experts

Documented overlays and fingerprints

Potential affordability

Built around devices farmers may already own

Field to multi-farm

Plant, field and farm identifiers

Potential sustainability

May reduce unnecessary inspections; to be tested

Impact statements are potential benefits, not proven outcomes.

Muskan Singh

Team Lead & AI / System Integration

Akash

Dataset & Annotation Research

Jeevith P

AI Model Development

Hitesh S L

Frontend & UX

Mahaida Hashmi

Backend, QA & Validation

PhytoTrace: Every Leaf Becomes Evidence

Problem	Similar leaf-mining symptoms make species-level and mixed-infestation diagnosis hard.
Solution	Smartphone-first support: detect, localize, explain and monitor.
Distinctive idea	Mixed-Infestation Evidence Map with calibrated uncertainty.
Approach	PWA + FastAPI + PyTorch / OpenCV; SQLite, PostgreSQL later; ONNX later.
Validation	Expert-labelled data, held-out splits, calibration, field trials.
Next steps	Licensed data; expert-confirmed Liriomyza and mixed samples; baseline models.

References

CHIP TO CROP 24-hour AgriTech Hackathon rulebook + problem statements

ICAR-Indian Institute of Horticultural Research, Bengaluru

EPPO Global Database — Tuta absoluta

Campos MR et al. (2017) Journal of Pest Science 90, 787–796

FastAPI documentation

PyTorch

ONNX Runtime

MDN — Progressive web apps

Datasets: to be identified and licence-checked (Phase 1).

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